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 *  History restored

Microsoft Windows [Version 10.0.26200.9711]
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C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp1 exp1.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp1.exe
Enter rows and columns: 3
3
Enter matrix elements:
0
0
5
0
8
0
3
0
0

Sparse Matrix (Triplet Representation):
Row     Column  Value
3       3       3
0       2       5
1       1       8
2       0       3

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp8

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 1
Enter roll number: 1
Enter name: anubhav
Enter position: 1

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 1
Enter roll number: 2
Enter name: aniker
Enter position: 2

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 4
Roll: 1 Name: anubhav
Roll: 2 Name: aniker

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice:
```

Problems Output Debug Console Terminal Ports

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp14 exp14.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp14
Enter number of vertices: 5
Enter number of edges: 4
Enter edges (u v):
0 1
0 2
1 3
1 4
Enter starting vertex: 0
DFS Traversal: 0 1 3 4 2
BFS Traversal: 0 1 2 3 4

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp14 exp14.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp14
Enter number of vertices: 5
Enter number of edges: 4
Enter edges (u v):
0 1
0 2
1 3
1 4
Enter starting vertex: 0
DFS Traversal: 0 1 3 4 2
BFS Traversal: 0 1 2 3 4

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>
```

```
3. Display
4. Exit
Enter choice: 2
Enter element to search: 30
Element found at index: 2

--- Hashing Operations ---
1. Insert
2. Search
3. Display
4. Exit
Enter choice: 3

Hash Table:
0 : 10
1 : 20
2 : 30
3 : Empty
4 : Empty
5 : Empty
6 : Empty
7 : Empty
8 : Empty
9 : Empty

--- Hashing Operations ---
1. Insert
2. Search
3. Display
4. Exit
Enter choice:
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp13   exp13.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp13

--- Hashing Operations ---
1. Insert
2. Search
3. Display
4. Exit
Enter choice: 1
Enter element: 10
Element inserted successfully!

--- Hashing Operations ---
1. Insert
2. Search
3. Display
4. Exit
Enter choice: 1
Enter element: 20
Element inserted successfully!

--- Hashing Operations ---
1. Insert
2. Search
3. Display
4. Exit
Enter choice: 1
Enter element: 30
Element inserted successfully!
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp12  exp12.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp12
Enter array size: 7
Enter array elements: 2
4
5
7
2
9
4

1. Linear Search
2. Binary Search
3. Exit
Enter choice: 1
Enter element to search: 5
Element found at position: 3
1. Linear Search
2. Binary Search
3. Exit
Enter choice: 2
Enter element to search: 4
Binary Search requires a sorted array!

1. Linear Search
2. Binary Search
3. Exit
Enter choice:
```

```
6. Heap Sort
7. Display Array
8. Exit
Enter choice: 6
Sorted using Heap Sort: 2 3 4 5 6 7 9

--- Sorting Algorithms ---
1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort
6. Heap Sort
7. Display Array
8. Exit
Enter choice: 7
2 3 4 5 6 7 9

--- Sorting Algorithms ---
1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort
6. Heap Sort
7. Display Array
8. Exit
Enter choice: 8
Exiting...
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>
```

```
8. Exit
Enter choice: 3
Sorted using Insertion Sort: 2 3 4 5 6 7 9

--- Sorting Algorithms ---
1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort
6. Heap Sort
7. Display Array
8. Exit
Enter choice: 4
Sorted using Merge Sort: 2 3 4 5 6 7 9

--- Sorting Algorithms ---
1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort
6. Heap Sort
7. Display Array
8. Exit
Enter choice: 5
Sorted using Quick Sort: 2 3 4 5 6 7 9

--- Sorting Algorithms ---
1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort
6. Heap Sort
7. Display Array
8. Exit
Enter choice: 6
Sorted using Heap Sort: 2 3 4 5 6 7 9
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp11 exp11.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp11
Enter number of elements: 7
Enter array elements: 2
5
7
3
9
6
4

--- Sorting Algorithms ---
1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort
6. Heap Sort
7. Display Array
8. Exit
Enter choice: 1
Sorted using Selection Sort: 2 3 4 5 6 7 9

--- Sorting Algorithms ---
1. Selection Sort
2. Bubble Sort
3. Insertion Sort
4. Merge Sort
5. Quick Sort
6. Heap Sort
7. Display Array
8. Exit
Enter choice: 2
Sorted using Bubble Sort: 2 3 4 5 6 7 9

--- Sorting Algorithms ---
1. Selection Sort
```

```
4. Preorder Traversal
5. Postorder Traversal
6. Exit
Enter choice: 1
Enter automobile type: ev
Enter company: tesla
Enter year of make: 2015

--- Binary Search Tree ---
1. Insert Automobile
2. Delete Automobile
3. Inorder Traversal
4. Preorder Traversal
5. Postorder Traversal
6. Exit
Enter choice: 3

Inorder Traversal:
sedan honda 1995
suv mahindra 2000
ev tesla 2015

--- Binary Search Tree ---
1. Insert Automobile
2. Delete Automobile
3. Inorder Traversal
4. Preorder Traversal
5. Postorder Traversal
6. Exit
Enter choice: 4

Preorder Traversal:
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp10 exp10.c

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp10

--- Binary Search Tree ---
1. Insert Automobile
2. Delete Automobile
3. Inorder Traversal
4. Preorder Traversal
5. Postorder Traversal
6. Exit
Enter choice: 1
Enter automobile type: sedan
Enter company: honda
Enter year of make: 1995

--- Binary Search Tree ---
1. Insert Automobile
2. Delete Automobile
3. Inorder Traversal
4. Preorder Traversal
5. Postorder Traversal
6. Exit
Enter choice: 1
Enter automobile type: suv
Enter company: mahindra
Enter year of make: 2000

--- Binary Search Tree ---
1. Insert Automobile
2. Delete Automobile
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp9 exp9.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp9
Create Binary Tree:
Enter value (-1 for no node): 10
Enter left child of 10
Enter value (-1 for no node): 20
Enter left child of 20
Enter value (-1 for no node): 30
Enter left child of 30
Enter value (-1 for no node): -1
Enter right child of 30
Enter value (-1 for no node): 25
Enter left child of 25
Enter value (-1 for no node): 95
Enter left child of 95
Enter value (-1 for no node): -1
Enter right child of 95
Enter value (-1 for no node): -1
Enter right child of 25
Enter value (-1 for no node): -1
Enter right child of 20
Enter value (-1 for no node): -1
Enter right child of 10
Enter value (-1 for no node): -1

Preorder Traversal: 10 20 30 25 95
Inorder Traversal: 30 95 25 20 10
Postorder Traversal: 95 25 30 20 10
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp7 exp7.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp7

--- Linear Queue Operations ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit
Enter your choice: 1
Enter value: 10
Element inserted successfully!

--- Linear Queue Operations ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit
Enter your choice: 1
Enter value: 20
Element inserted successfully!

--- Linear Queue Operations ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit
Enter your choice: 2
Deleted element: 10

--- Linear Queue Operations ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit
Enter your choice: 3
Queue elements: 20

--- Linear Queue Operations ---
1. Insert (Enqueue)
2. Delete (Dequeue)
3. Display
4. Exit
Enter your choice:
```

```
Enter your choice: 3
Top element: 30

--- Stack Operations ---
1. Push
2. Pop
3. Peek
4. Traverse
5. Exit
Enter your choice: 4
Stack elements: 30 20 10

--- Stack Operations ---
1. Push
2. Pop
3. Peek
4. Traverse
5. Exit
Enter your choice: 2
Popped element: 30

--- Stack Operations ---
1. Push
2. Pop
3. Peek
4. Traverse
5. Exit
Enter your choice: 4
Stack elements: 20 10

--- Stack Operations ---
1. Push
2. Pop
3. Peek
4. Traverse
5. Exit
Enter your choice:
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp6 exp6.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp6

--- Stack Operations ---
1. Push
2. Pop
3. Peek
4. Traverse
5. Exit
Enter your choice: 1
Enter value: 10
Element pushed successfully!

--- Stack Operations ---
1. Push
2. Pop
3. Peek
4. Traverse
5. Exit
Enter your choice: 1
Enter value: 20
Element pushed successfully!

--- Stack Operations ---
1. Push
2. Pop
3. Peek
4. Traverse
5. Exit
Enter your choice: 1
Enter value: 30
Element pushed successfully!

--- Stack Operations ---
1. Push
2. Pop
3. Peek
```

```
3. Pop Stack 1
4. Pop Stack 2
5. Display Stack 1
6. Display Stack 2
7. Exit
Enter choice: 2
Enter value: 200
Element pushed into Stack 2

1. Push Stack 1
2. Push Stack 2
3. Pop Stack 1
4. Pop Stack 2
5. Display Stack 1
6. Display Stack 2
7. Exit
Enter choice: 5
Stack 1: 20 10

1. Push Stack 1
2. Push Stack 2
3. Pop Stack 1
4. Pop Stack 2
5. Display Stack 1
6. Display Stack 2
7. Exit
Enter choice: 6
Stack 2: 200 100

1. Push Stack 1
2. Push Stack 2
3. Pop Stack 1
4. Pop Stack 2
5. Display Stack 1
6. Display Stack 2
7. Exit
Enter choice:
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp5 exp5.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp5

1. Push Stack 1
2. Push Stack 2
3. Pop Stack 1
4. Pop Stack 2
5. Display Stack 1
6. Display Stack 2
7. Exit
Enter choice: 1
Enter value: 10
Element pushed into Stack 1

1. Push Stack 1
2. Push Stack 2
3. Pop Stack 1
4. Pop Stack 2
5. Display Stack 1
6. Display Stack 2
7. Exit
Enter choice: 1
Enter value: 20
Element pushed into Stack 1

1. Push Stack 1
2. Push Stack 2
3. Pop Stack 1
4. Pop Stack 2
5. Display Stack 1
6. Display Stack 2
7. Exit
Enter choice: 2
Enter value: 100
Element pushed into Stack 2

1. Push Stack 1
```

```
3. Display
4. Exit
Enter choice: 1
Enter College ID: 3
Enter College Name: bpit
College inserted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 3

College Details:
College ID: 3, College Name: bpit
College ID: 2, College Name: vips
College ID: 1, College Name: adgips

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 2
College deleted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 3

College Details:
College ID: 3, College Name: bpit
College ID: 2, College Name: vips

1. Insert at Front
2. Delete at End
3. Display
4. Exit
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp4 exp4.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp4

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 1
Enter College ID: 1
Enter College Name: adgips
College inserted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 1
Enter College ID: 2
Enter College Name: vips
College inserted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 1
Enter College ID: 3
Enter College Name: bpit
College inserted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 3

College Details:
```

Problems Output Debug Console Terminal Ports

```
Enter Employee ID: 3
Enter Employee Name: aditya
Enter Salary: 100000
Employee inserted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 3

Employee Details:
ID: 3, Name: aditya, Salary: 100000
ID: 2, Name: aniket, Salary: 150000
ID: 1, Name: anubhav, Salary: 200000

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 2
Employee deleted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 3

Employee Details:
ID: 3, Name: aditya, Salary: 100000
ID: 2, Name: aniket, Salary: 150000

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 4
Exiting...
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>
```

```
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C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>g++ -o exp3 exp3.cpp

C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp3

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 1
Enter Employee ID: 1
Enter Employee Name: anubhav
Enter Salary: 200000
Employee inserted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 1
Enter Employee ID: 2
Enter Employee Name: aniket
Enter Salary: 150000
Employee inserted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 1
Enter Employee ID: 3
Enter Employee Name: aditya
Enter Salary: 100000
Employee inserted successfully!

1. Insert at Front
2. Delete at End
3. Display
4. Exit
Enter choice: 3

Employee Details:
ID: 3, Name: aditya, Salary: 100000
ID: 2, Name: aniket, Salary: 150000
```

```
Student Details:
Roll No: 3, Name: aditya
Roll No: 2, Name: aniket
Roll No: 1, Name: anubhav

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 2
Enter roll number to delete: 1
Student deleted successfully!

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 4

Student Details:
Roll No: 3, Name: aditya
Roll No: 2, Name: aniket

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 5
Exiting...
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>
```

```
Enter choice: 4

Student Details:
Roll No: 1, Name: anubhav
Roll No: 2, Name: aniket
Roll No: 3, Name: aditya

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 3
List reversed successfully!

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 4

Student Details:
Roll No: 3, Name: aditya
Roll No: 2, Name: aniket
Roll No: 1, Name: anubhav

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 2
Enter roll number to delete: 1
Student deleted successfully!

1. Insert Student
2. Delete Student
3. Reverse List
```

```
C:\Programming Practice\PROJECTS\Programming Practice\DS LAB>.\exp2

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 1
Enter roll number: 1
Enter student name: anubhav
Enter position: 1

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 1
Enter roll number: 2
Enter student name: aniket
Enter position: 2

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 1
Enter roll number: 3
Enter student name: aditya
Enter position: 3

1. Insert Student
2. Delete Student
3. Reverse List
4. Display List
5. Exit
Enter choice: 4
```